SOLVING QUADRATIC EQUATIONS

BASIC ALGEBRA

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Rational Expression

Standard Form

Quadratic Formula

RATIONAL EXPRESSION

Simplify $\frac{21}{45}$ by reducing to **lowest terms**

$$\frac{\cancel{3} \cdot 7}{\cancel{3} \cdot 15}$$

A mathematical expression that can be represented in a **fractional form**

$$\boxed{\frac{7}{15}}$$

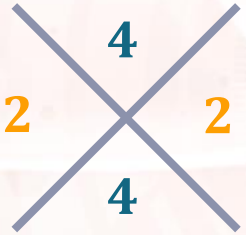
ans

Simplify $\dfrac{3x+6}{x^2+4x+4}$

by reducing to **lowest terms**

GFC ,

$$\frac{3x+6}{3} = x+2$$



$$\frac{3\cancel{(x+2)}}{(x+2)\cancel{(x+2)}}$$

$$\boxed{\frac{3}{(x+2)}}$$

ans

Compute $\frac{4}{3} \cdot \frac{2}{5}$

$$\frac{4 \cdot 2}{3 \cdot 5}$$

$$\boxed{\frac{8}{15}}$$

ans

Compute $\dfrac{\frac{4}{5}}{\frac{2}{3}}$

$$\frac{4}{5} \cdot \frac{3}{2}$$

$$\frac{12}{10}$$

$$\frac{\cancel{2} \cdot 6}{\cancel{2} \cdot 5}$$

$$\boxed{\frac{6}{5}}$$

ans

Compute $\dfrac{\frac{x^2+x}{x+4}}{\frac{x+1}{x^2-16}}$

$$\frac{x^2+x}{x+4} \cdot \frac{x^2-16}{x+1}$$

$$\frac{x(x+1)}{x+4} \cdot \frac{(x+4)(x-4)}{x+1}$$

GCF ,

$$\frac{x^2+x}{x} = x+1$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$x^2 - 4^2 = (x-4)(x+4)$$

Compute $\dfrac{\frac{x^2+x}{x+4}}{\frac{x+1}{x^2-16}}$

$$\frac{x^2+x}{x+4} \cdot \frac{x^2-16}{x+1}$$

GCF ,

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$$\frac{x\cancel{(x+1)}}{\cancel{x+4}} \cdot \frac{\cancel{(x+4)}(x-4)}{\cancel{x+1}}$$

$$\boxed{x(x-4)}$$

ans

Subtract $\frac{7}{6} - \frac{4}{15}$

CD ,

$6 \cdot 15 = 90$

$\frac{7}{6} \cdot \frac{15}{15} - \frac{4}{15} \cdot \frac{6}{6}$

$\frac{105}{90} - \frac{24}{90}$

$\frac{81}{90}$

$\frac{\cancel{9} \cdot 9}{\cancel{9} \cdot 10}$

$\boxed{\frac{9}{10}}$

ans

Subtract $\frac{7}{6} - \frac{4}{15}$

LCD ,

$6 = 2 \cdot 3$

$15 = 3 \cdot 5$

$\longrightarrow$ $2 \cdot 3 \cdot 5$

$$\frac{7}{2 \cdot 3} \cdot \frac{5}{5} - \frac{4}{3 \cdot 5} \cdot \frac{2}{2}$$

$$\frac{35}{30} - \frac{8}{30}$$

$$\frac{27}{30}$$

$$\frac{\cancel{3} \cdot 9}{\cancel{3} \cdot 10}$$

$$\boxed{\frac{9}{10}}$$

ans

Add $\frac{3}{2x+2} + \frac{5}{x^2-1}$

LCD ,

$2x + 2 = 2(x + 1)$

$x^2 - 1 = (x + 1)(x - 1)$

$\longrightarrow$ $2(x + 1)(x - 1)$

$$\frac{3}{2(x+1)} \cdot \frac{(x-1)}{(x-1)} + \frac{5}{(x+1)(x-1)} \cdot \frac{2}{2}$$

$$\frac{3(x-1)}{2(x+1)(x-1)} + \frac{10}{2(x+1)(x-1)}$$

$$\frac{3(x-1)+10}{2(x+1)(x-1)}$$

$$\frac{3x-3+10}{2(x+1)(x-1)}$$

$$\boxed{\frac{3x+7}{2(x+1)(x-1)}}$$

ans

STANDARD FORM

STANDARD FORM

A quadratic equation is a **second**-degree polynomial equation

$$ax^2 + bx + c = 0$$

Find all **real solutions** for the equation

$$y^2 = 18 - 7y$$

$$y^2 + 7y - 18 = 0$$

$$(y + 9)(y - 2) = 0$$

$$(y + 9) = 0 \qquad (y - 2) = 0$$

$$\boxed{y = -9} \qquad \boxed{y = 2}$$

ans *ans*

Check ,

$$-9^2 = 18 - 7(-9) \qquad 2^2 = 18 - 7(2)$$

$$81 = 18 + 63 \qquad 4 = 18 - 14$$

$$81 = 81 \qquad 4 = 4$$

Find all **real solutions** for the equation

$$w^2 = 121$$

$$w^2 - 121 = 0$$

$$w^2 - 11^2 = 0$$

$$(w + 11)(w - 11) = 0$$

$$(w + 11) = 0 \qquad (w - 11) = 0$$

$$\boxed{w = -11} \qquad \boxed{w = 11}$$

ans *ans*

$$w^2 = 121$$

$$w = \pm\sqrt{121}$$

$$\boxed{w = \pm 11}$$

ans

Find all **real solutions** for the equation

$$x(x+2)=7$$

$$x^2+2x=7$$

$$x^2+2x-7=0$$

$$\begin{array}{c} -7 \\ \times \\ 2 \end{array}$$

QUADRATIC FORMULA

QUADRATIC FORMULA

The **solutions** to the quadratic equation

are given by the formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Find all **real solutions** for the equation

$$x(x+2)=7$$

$$x^2+2x=7$$

$$x^2+2x-7=0 \qquad ax^2+bx+c=0$$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$a=1 \qquad b=2 \qquad c=-7$$

$$x=\frac{-2\pm}{2(1)}$$

Find all **real solutions** for the equation

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$$x^2+2x=7$$

$$x^2+2x-7=0 \qquad ax^2+bx+c=0$$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$a=1 \qquad b=2 \qquad c=-7$$

$$x=\frac{-2\pm\sqrt{2^2-4(1)(-7)}}{2(1)}$$

$$x=\frac{-2\pm\sqrt{32}}{2}$$

$$x=\frac{-2\pm\sqrt{16\cdot 2}}{2}$$

$$x=\frac{-2\pm\sqrt{16}\cdot\sqrt{2}}{2}$$

$$x=\frac{-2\pm 4\cdot\sqrt{2}}{2}$$

$$x=-1\pm 2\sqrt{2}$$

$$\boxed{x=-1+2\sqrt{2}} \qquad \boxed{x=-1-2\sqrt{2}}$$

ans *ans*

Find all **real solutions** for the equation

$$\frac{1}{2}y^2 = \frac{1}{3}y - 2$$

$$\frac{1}{2}y^2 - \frac{1}{3}y + 2 = 0$$

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$$6\left(\frac{1}{2}y^2 - \frac{1}{3}y + 2 = 0\right)$$

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$$a = 3 \qquad b = -2 \qquad c = 12$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(3)(12)}}{2(3)}$$

Find all **real solutions** for the equation

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$$x = \frac{2 \pm \sqrt{-140}}{6}$$

$\boxed{\text{No real solution}}$

ans

SEATWORK